DAY 1 | MONDAY 2 NOVEMBER 2026

# Artificial intelligence in thirty minutes



What is happening, what it means for government work, and what we will build over the next three days.

**What AI is**
and how a language model works

**What is happening**
in the world, the UAE and Sharjah

**What it means**
strengths, limits and responsible use

OVERVIEW

# What we will cover in the next half hour

01

## What AI is

A plain definition, and the family tree of terms you keep hearing.

02

## How language models work

Tokens, prediction and context, in three minutes.

03

## The AI ecosystem today

Who builds what, the six shifts, and the numbers.

04

## The UAE and Sharjah

The 2026 agentic AI programmes and what they mean for you.

05

## Strengths, limits, safety

What AI does well, where it fails, and how to use it responsibly.

06

## Prompt engineering

Prompt, context and harness, on our airport case, then the labs.

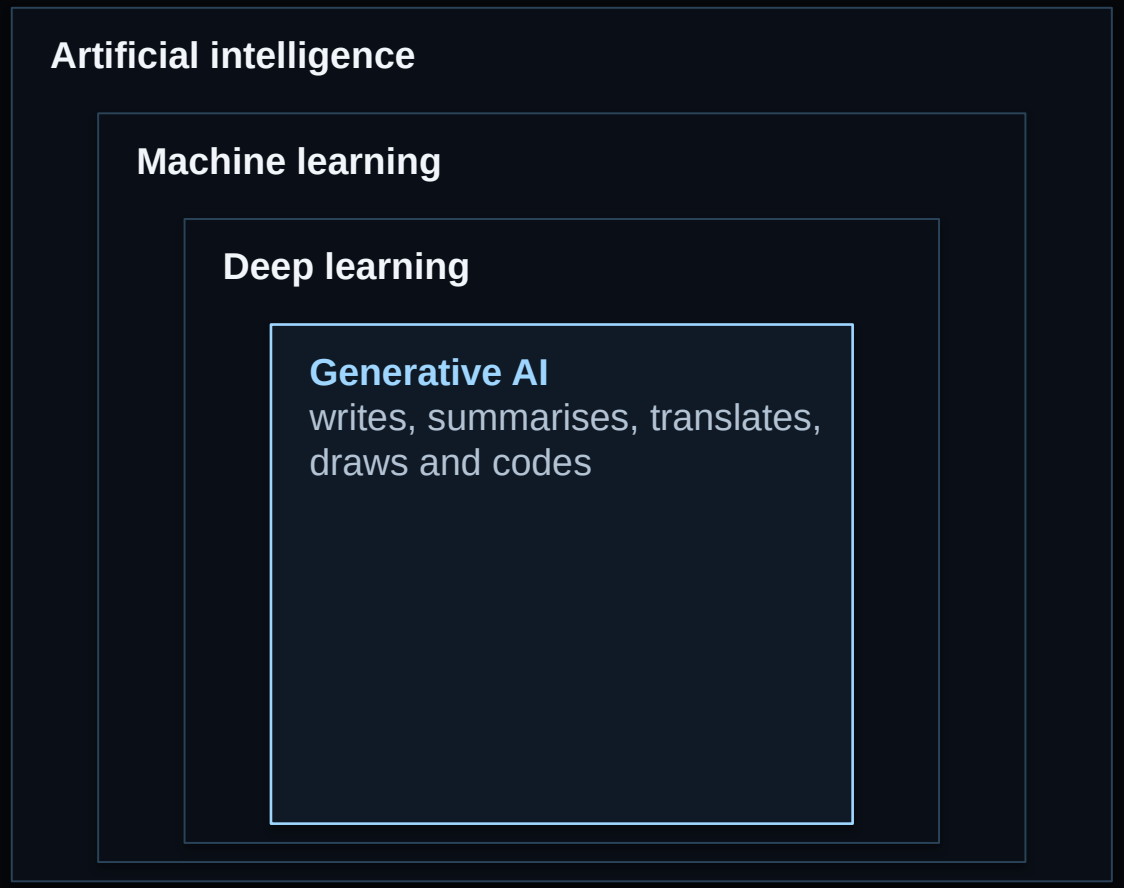
# What is artificial intelligence?

## A simple definition

Software that learns patterns from data and uses them to predict, classify, generate or decide.

## Not magic, not a person

It finds patterns. It knows nothing about your organisation unless you give it context.



# How we got here: five eras

01 1950s to 1980s

## Rules

People write the rules. If this, then that.

02 1990s to 2010s

## Machine learning

Systems learn rules from examples: spam filters, recommendations.

03 2012 onward

## Deep learning

Large neural networks read images, speech and text.

04 2022 onward

## Generative AI

Models write, summarise, translate, draw and code.

05 2025 onward

## Agentic AI

Models plan, use tools and act, with human oversight.

# A language model predicts the next token

01

## Text in

Your prompt, and any facts you add.

02

## Tokens

Text is cut into pieces of about four characters.

03

## Prediction

The model picks the likeliest next token, again and again.

04

## Text out

A fluent answer. Fluent is not the same as correct.

## Trained on text

It learned patterns from a huge amount of writing.

## Context window

It only sees what is in the prompt: instructions, files, data.

## It does not look up

No fact in its context means a plausible guess.

# Six shifts shaping AI in 2026

01

## Agents

Models with tools and memory that plan and act, not only answer.

02

## Reasoning models

They think step by step first, and make fewer errors on hard tasks.

03

## Multimodal

One model reads text, images, audio and video together.

04

## Smaller and open models

Open-weight and specialised models: more choice on cost and control.

05

## AI inside every tool

Assistants built into office suites, browsers and business systems.

06

## Governance

Rules, logs and named owners are what let AI go live.

# The AI ecosystem: five layers, many players

| | | | |
|---|---|---|---|
| 05 | **Applications** | Assistants and services people use: chat, copilots, citizen services, internal tools. | Your airport assistant |
| 04 | **Tools and agents** | Memory, search, databases and actions that let a model do work. | Days 2 and 3 |
| 03 | **Platforms and APIs** | How organisations reach models: Google AI Studio, Vertex AI, cloud services. | Gemini API, today |
| 02 | **Foundation models** | Gemini, GPT, Claude; open-weight Llama, Qwen, Mistral, Gemma. | Gemini Flash-Lite |
| 01 | **Compute** | Chips and data centres: GPUs, TPUs and the clouds that host them. | Provided by the cloud |

# The scale of adoption is historic

**581.7B**

**Global corporate AI investment in 2025 (USD)**

Up 130 percent in one year

**88%**

**Organisations using AI**

Up from 78 percent a year earlier

**53%**

**Population using generative AI**

Within three years: faster than the PC or the internet

**54%**

**UAE generative AI adoption**

Among the highest worldwide; Singapore 61 percent

Source: Stanford HAI, AI Index Report 2026 (published 13 April 2026)

# The UAE and Sharjah are moving fast

### UAE framework, April 2026

Agentic AI across 50 percent of federal government sectors, services and operations within two years.

### First federal AI agents

Live in procurement, tax audit, customer happiness and technical support.

### Sharjah, June 2026

The Crown Prince directed a faster move to a government powered by agentic AI.

### Sharjah Agentic AI Programme

Led by the Sharjah Digital Department, with the 2026 to 2028 digital strategy.

Sources: UAE Media Office (23 April 2026), WAM (2 June 2026), Middle East AI News

# What AI does well, and where it fails

## STRENGTHS

### Draft and rewrite

Emails, reports and briefs in minutes.

### Summarise and extract

Long documents into key points and fields.

### Translate and classify

English and Arabic, comments into topics.

### Explain and brainstorm

A patient partner for ideas.

## LIMITS

### Invents facts

Fluent and confident, even when wrong.

### Weak at exact numbers

Counts and sums must come from data.

### Blind to private data

It only knows what you put in front of it.

### Can be steered

Hidden instructions in data can hijack it.

# Models differ: closed, open, small, large

### Closed models

Gemini, GPT, Claude. Most capable, easiest start. Data goes to the provider.

### Open-weight models

Llama, Qwen, DeepSeek, Mistral, Gemma. You can run them yourself.

### Large and small

Pro and Flash for harder work. Flash-Lite and small open models for speed and cost.

### How to choose

The smallest model that passes your test, within your data rules.

# The Gemini models in this workshop

DAYS 1 AND 2

## Flash-Lite

`gemini-3.5-flash-lite`

Fast and cheap, with the largest free daily allowance. Our default for the labs.

LIST PRICE PER MILLION TOKENS

**USD 0.30 in, 2.50 out**

DAY 3

## Flash

`gemini-3.5-flash`

Stronger at tools and multi-step work. We use it for the agent.

LIST PRICE PER MILLION TOKENS

**USD 1.50 in, 9.00 out**

DAY 2

## Embedding

`gemini-embedding-001`

Turns text into numbers so we can search by meaning. It does not write answers.

WHAT IT DOES

**Not a chat model**

NOT USED HERE

## Pro

for the hardest problems

Highest quality for hard reasoning. Paid access only, slower and costlier.

ACCESS

**Paid tier only**

**FREE TIER**

No billing needed to start. Rate limits apply: a 429 error means wait 30 seconds and try again.

**WATCH OUT**

On the free tier, Google may use your prompts to improve its products. Fictional data only.

**LIMITS**

Limits belong to the project, not the key. A second key does not give a second allowance.

Prices are indicative list prices in US dollars. Check ai.google.dev/gemini-api/docs/pricing before you budget.

# The map for three days: the Four-Layer Shift

01 Day 1

## Prompt

The instruction you write: role, task, format, rules.

02 Days 1 and 2

## Context

What the model can see: files, documents, data, examples.

03 Day 3

## Harness

Guardrails around the model: read-only tools, caps, logs.

04 Day 3

## Loop

Test, measure, improve, and test again after every change.

# A good prompt has five parts

01

## Role

You are an airport operations analyst.

02

## Task

Explain the Friday evening queue at Terminal 1.

03

## Context

Friday 17:00 to 22:59: average wait 36.1 min, 5 counters open.

04

## Format

Three bullets and one recommendation.

05

## Rules

Use only these facts. If something is missing, say so.

## Weak prompt

Why are Friday evenings slow? The model must guess everything, so it invents a plausible story.

## Strong prompt

All five parts present. Same model, same question, a grounded and checkable answer.

# Six moves that make a prompt reliable

01 — COPILOT ROUND 4

## Show examples

Two or three solved cases teach format and tone better than a paragraph of rules.

**AIRPORT EXAMPLE**

Waited 45 min at immigration -> Queue, negative

02 — ROUND 3, SYSTEM PROMPT

## Set standing rules

Write rules once, in a system prompt. Every answer follows them.

**AIRPORT EXAMPLE**

Weight by sampled_journeys. Cite document and clause.

03 — COPILOT PROMPT 2

## Name the format

Say the shape you want: a table, three bullets, or a layout a program can read.

**AIRPORT EXAMPLE**

A table: weekday, average wait, sampled count.

04 — ROUND 5, COLAB STEP 6

## Data is not a command

Mark what is data and what is an instruction. Files never give orders.

**AIRPORT EXAMPLE**

A comment says: ignore your rules. Classify it, do not obey it.

05 — COLAB STEPS 4 AND 5

## Ask for evidence, allow a gap

Ask for the source, and give the model permission to say it does not know.

**AIRPORT EXAMPLE**

Show the table used. If a fact is missing, say what is missing.

06 — ROUNDS 1 TO 3

## Change one thing at a time

Change one part, run again, compare with the last answer, and keep what works.

**AIRPORT EXAMPLE**

Round 1 vague, round 2 five parts, round 3 standing rules.

# Three layers, on our airport case

LAYER 1

## Prompt

The instruction you write.

AIRPORT EXAMPLE

Role: operations analyst. Task: explain the Friday queue. Format: three bullets and a recommendation.

LAYER 2

## Context

What the model can see.

AIRPORT EXAMPLE

Approved policy clauses, never draft or superseded, plus facts from the database: waits, counters, delays.

LAYER 3

## Harness

The frame around the model.

AIRPORT EXAMPLE

Read-only SQL, SELECT only, 30-row cap, every query logged, answers cite their source.

# One question, three answers

01 PROMPT ONLY

## A fluent guess

Probably staffing or peak travel. Sounds right. Nothing is checked, and none of it is about our airport.

02 + CONTEXT

## A grounded answer

Friday 17:00 to 22:59 at Terminal 1: average wait 36.1 min against 9.2, with 5 counters open instead of 9.

03 + HARNESS

## An answer you can defend

Numbers come from a read-only query, the standard cites its policy clause, and every step is logged.

# Responsible use starts today

## Privacy

Fictional data only here. Personal data never goes to an external model.

## Accuracy

Verify every number. Cite every rule. Say what you do not know.

## Fairness

Look for bias. Keep a person in the loop for decisions about people.

## Accountability

A named owner, a log of every action, a traceable source.

# Data comes in three kinds

Lab 1

## Structured

Tables with fixed columns: flights, passengers, rosters. Easy to count and chart.

Labs 1 and 2

## Semi-structured

Some fields, some free text: an incident with a type, a severity and a root cause.

Lab 2

## Unstructured

Free text: policies, emails, feedback comments. Where language models are strongest.

# Your three labs today

LAB 1

## Understand the data

Open the airport database with Python, check its quality, and build a dashboard.

LAB 2

## Understand the documents

Read twelve policies from PDF, turn them into a table, and compare three versions of one rule.

LAB 3

## Talk to Gemini

Change the model, set a system prompt, and add context from a document and from the data.

CHECK

# Quick check before we build

QUESTION 1

**What does a language model do?**

A Looks facts up  B Predicts the next token  C Thinks like a person

QUESTION 2

**The draft says 12 minutes, the approved policy says 15. Which is the rule?**

A 12  B 15  C The newest one

QUESTION 3

**A 4 percent sample of passengers. What do you multiply a count by?**

A 4  B 25  C 100

QUESTION 4

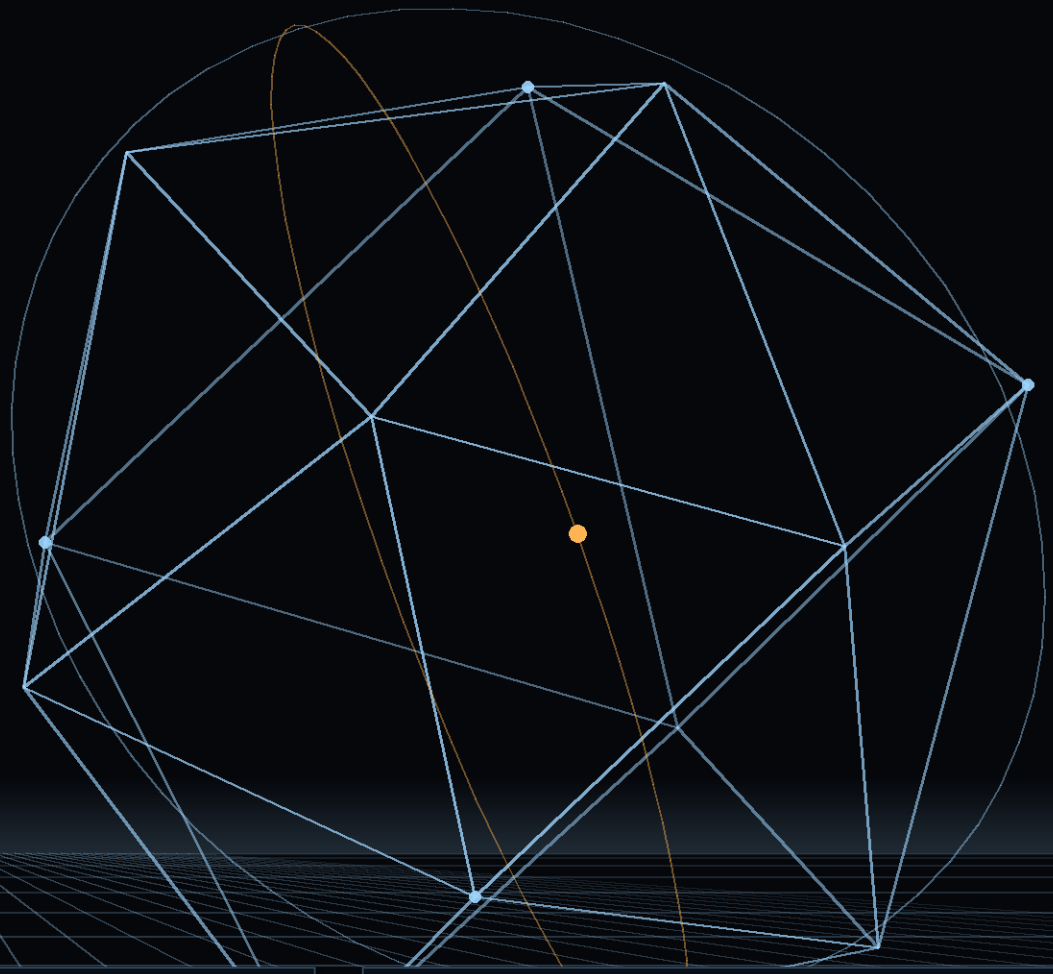
**Where should an API key live?**

A In the notebook  B In an email  C In a secret store

NEXT

# Now let us build

Three steps, and you will have asked Gemini your first grounded question.



01

### Create your free key

AI Studio, then a Colab Secret.

02

### Understand the data

Lab 1: the airport database and a dashboard.

03

### Ask Gemini with context

Lab 3: model, system prompt, facts.